Scale resolved analysis of brain functional connectivity networks with spectral entropies

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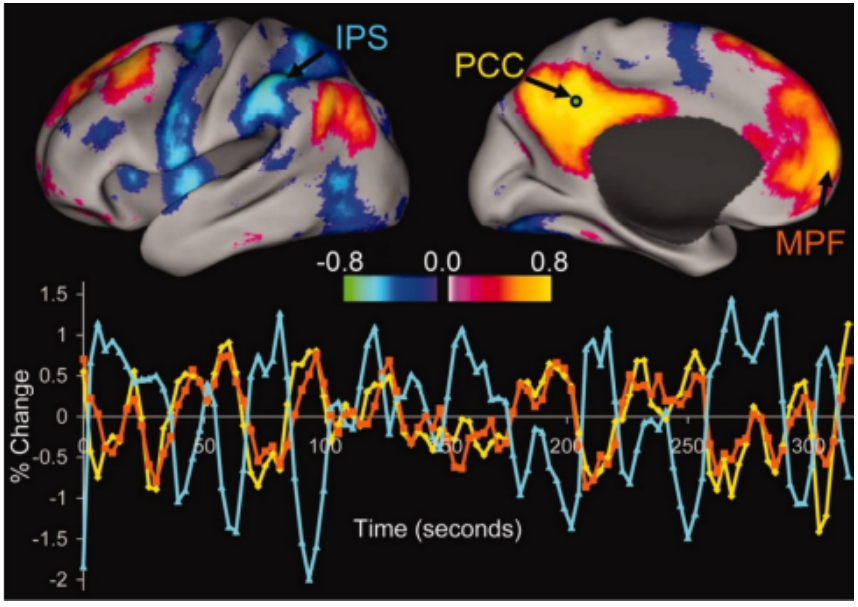
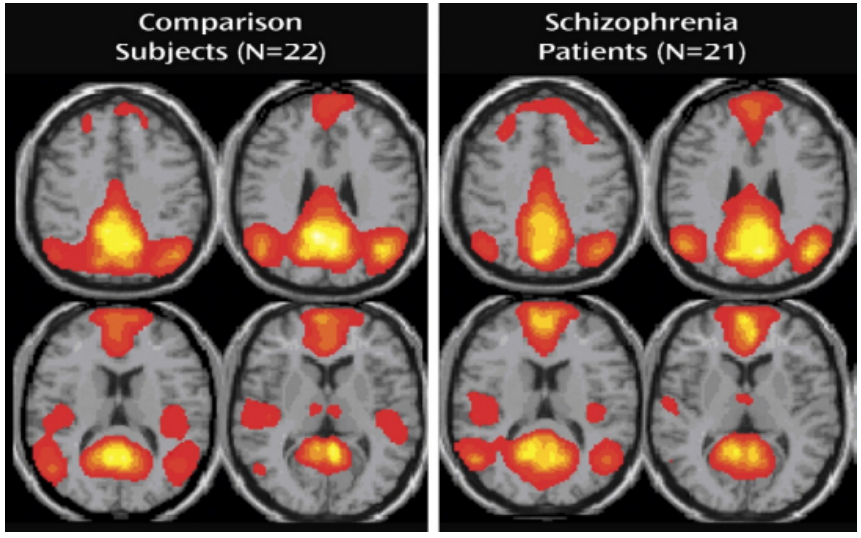
Lipari, July 15, 2019

Resting state brain networks



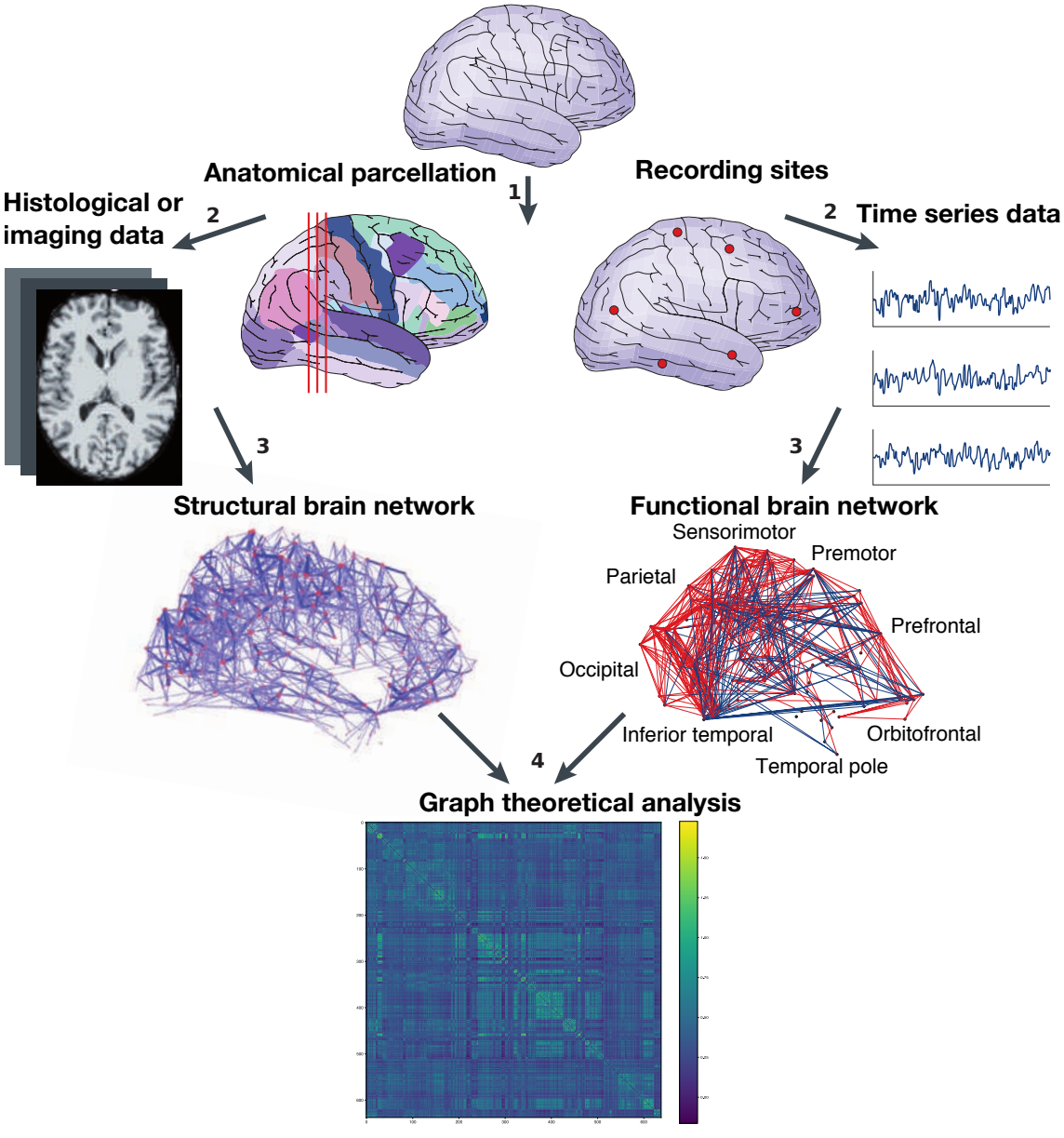
Fox et al. 2005



Greicius et al. 2007

- Spontaneuos fluctuations of BOLD fMRI signal
- Subject at rest, 5 minutes.
- Effects of motion.
- Effects of thresholding.
- Effects of diseases.

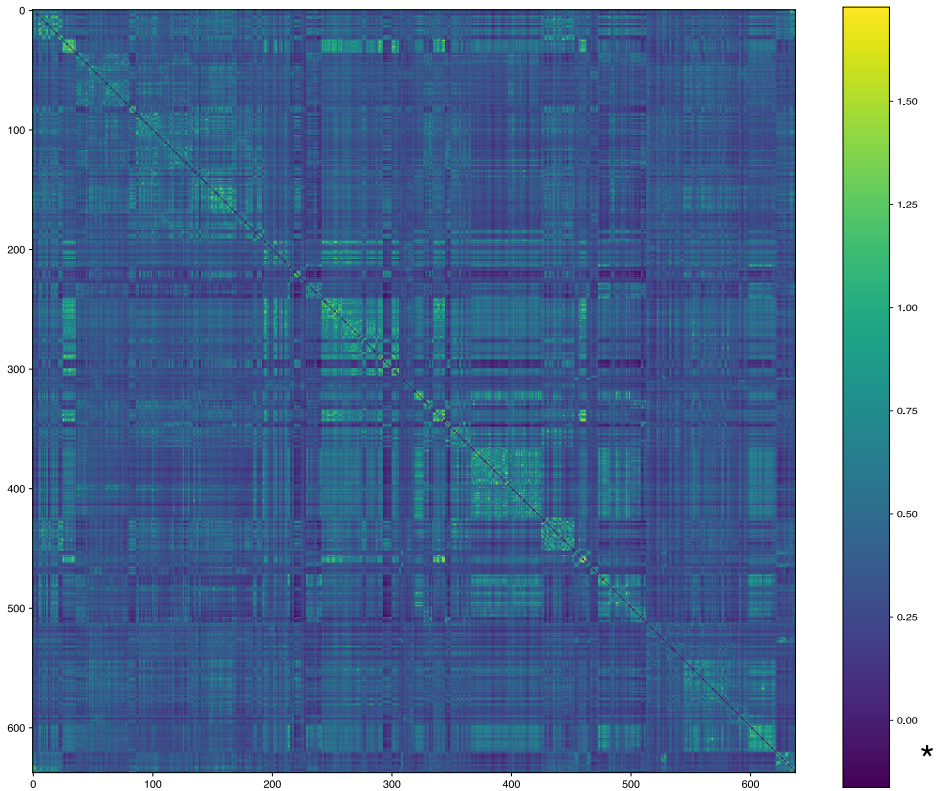
Graph theory to access information in rs-fMRI

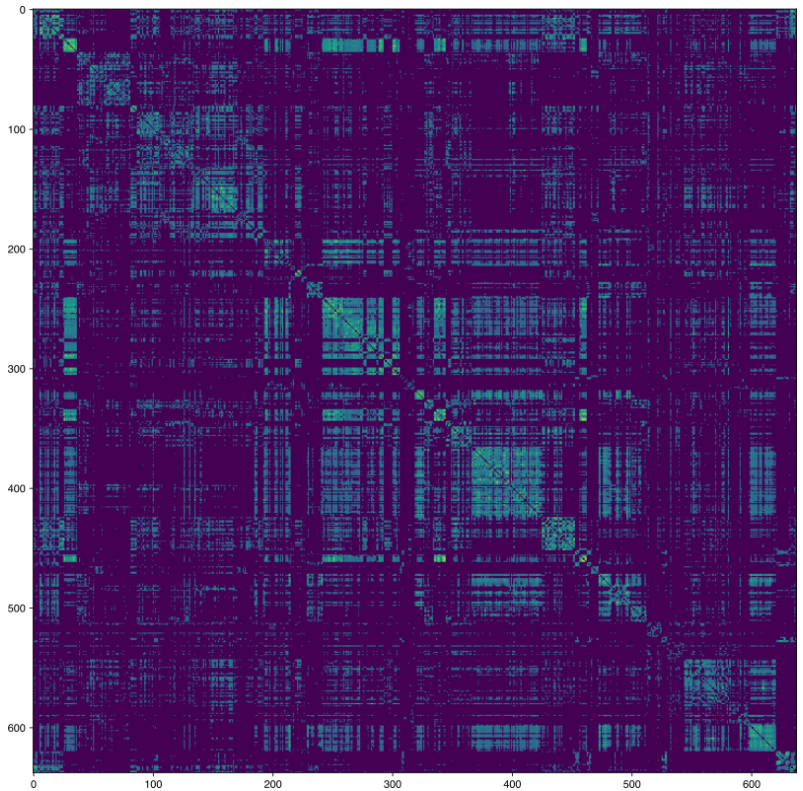


Continuous association measure (correlation, mutual information, coherence etc...)

How to compare brain networks?

- Weights are positive and negative.
- Matrices are dense (computationally hard).
- Spurious correlations (e.g. motion).





Before thresholding

After thresholding

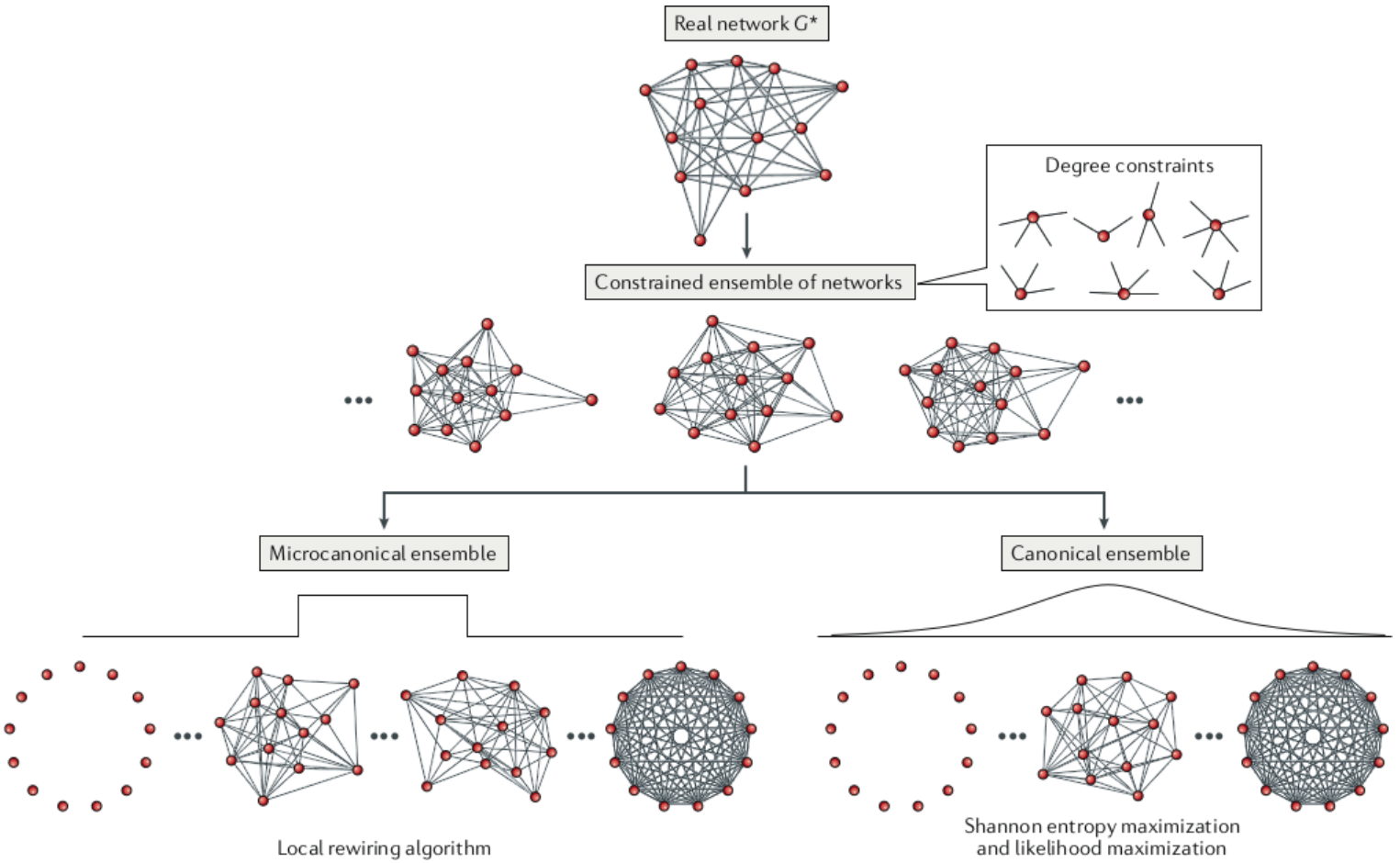
Appropriate null models are needed

MaxEnt framework helps

- Maximally non-committal to missing information.
- How far are networks from random?
- Detection of significant structural patterns in real networks.

- Maximum Entropy Random Graphs (classical)
- Spectral entropies (quantum)

Probability distributions on graphs



Maximize entropy $S(\mathcal{G}) = -\sum_{G \in \mathcal{G}} P(G) \ln P(G)$ with constraints.

$$P(G) = \frac{e^{-H}}{Z}$$

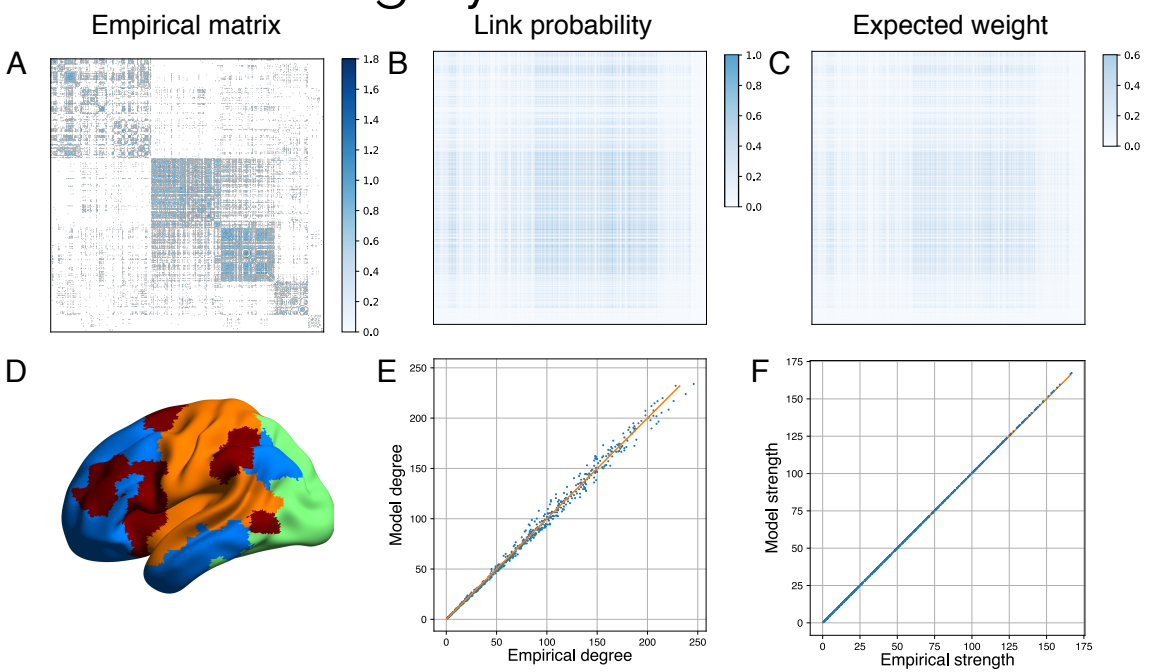
Continuous Enhanced Configuration model

- Degree sequence $k_i = \sum_j \Theta(w_{ij} - t)$
- Strength sequence $s_i = \sum_j w_{ij}\Theta(w_{ij} - t)$

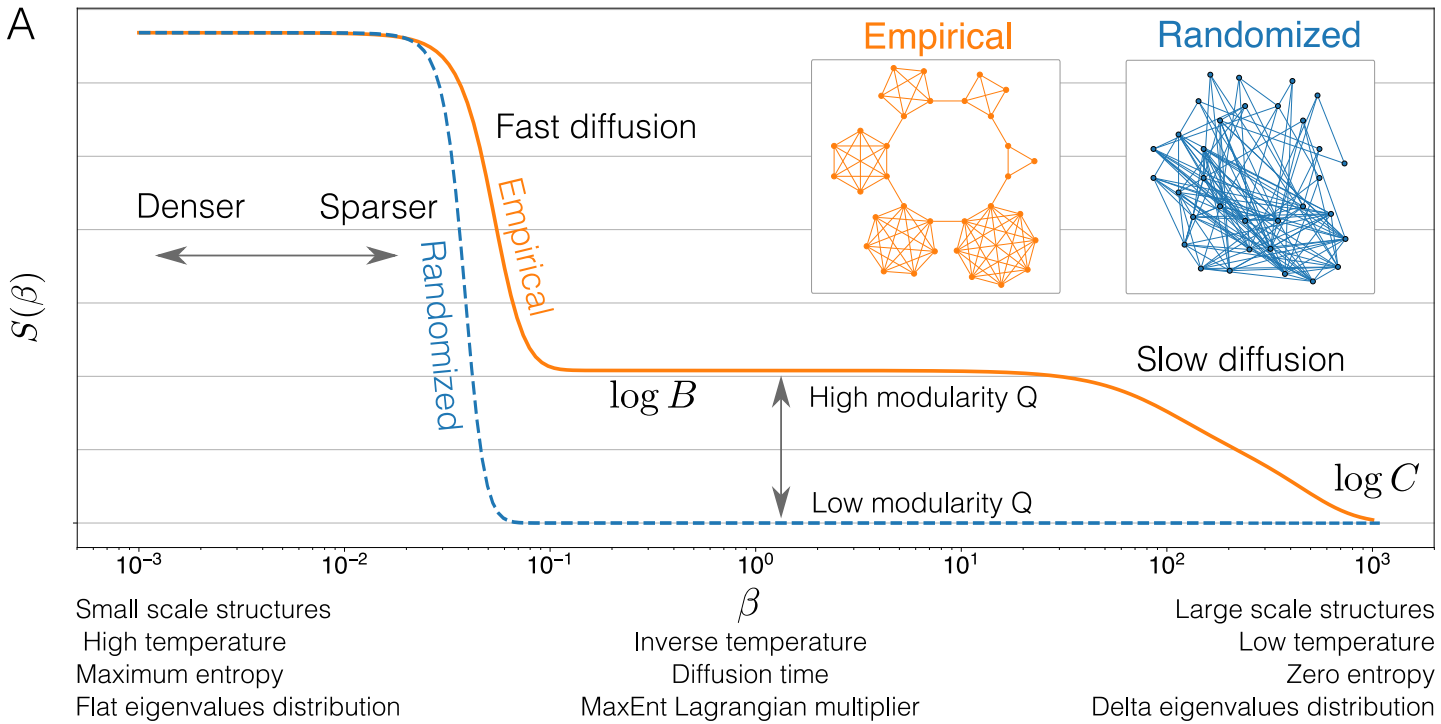
Hamiltonian: 2n parameters

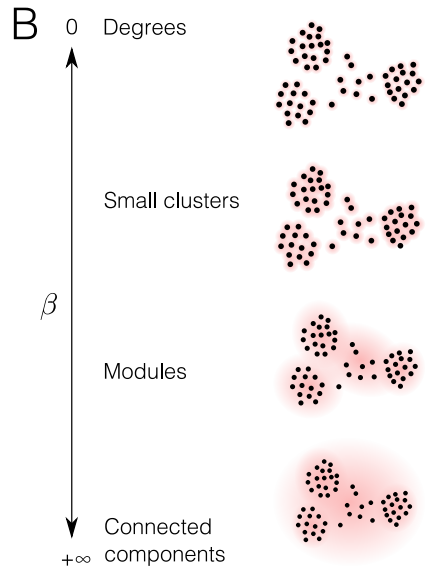
$$H(G|\boldsymbol{\alpha}, \boldsymbol{\beta}) = \sum_{i<j} \alpha_{ij}\Theta(w_{ij} - t) + \beta_{ij} w_{ij}\Theta(w_{ij} - t)$$

Model fitting by maximum likelihood.



Spectral entropies and diffusion on networks





Von Neumann entropy and density

$$S(\boldsymbol{\rho}) = -\operatorname{Tr}[\boldsymbol{\rho}\ln\boldsymbol{\rho}] \qquad \boldsymbol{\rho} = \frac{e^{-\beta\mathbf{L}}}{\operatorname{Tr}[e^{-\beta\mathbf{L}}]}$$

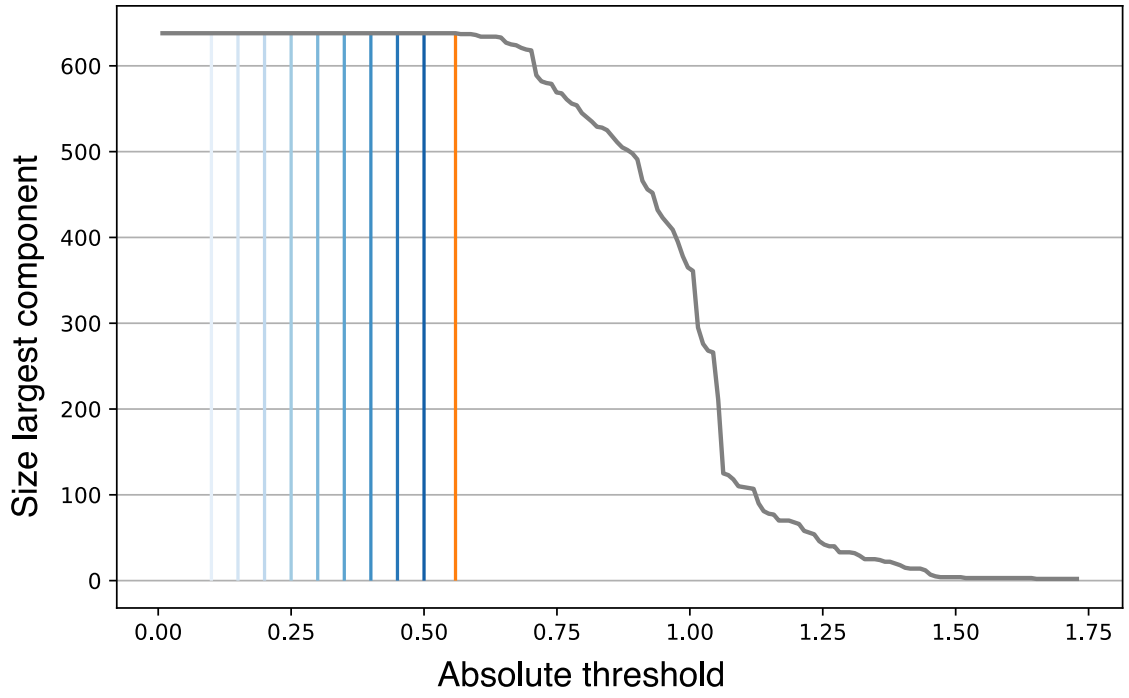
Shaped by Laplacian eigenvalues (spectral gap)

Von Neumann relative entropy: **scale resolved network comparison**

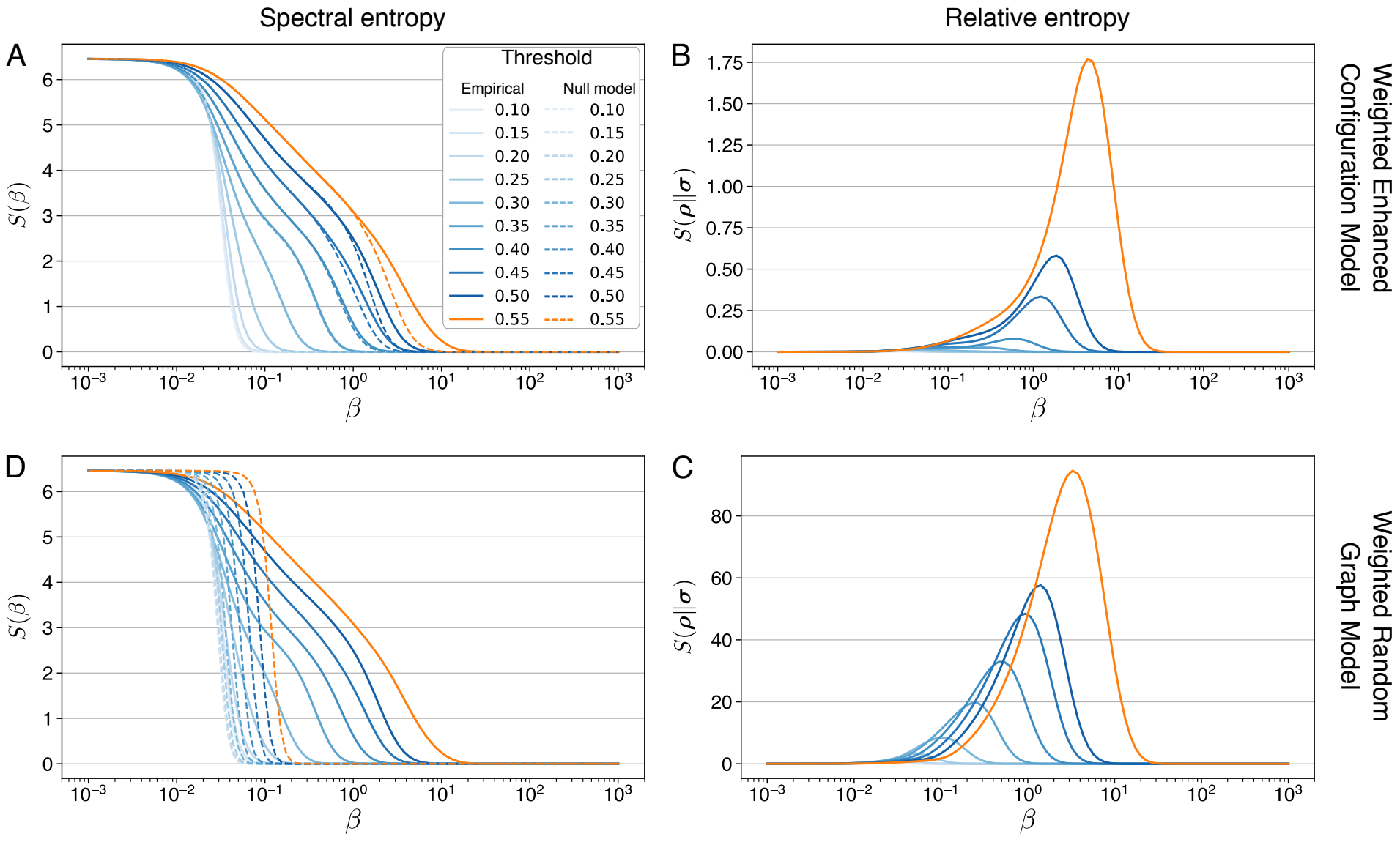
$$S(\boldsymbol{\rho}\|\boldsymbol{\sigma}) = \operatorname{Tr}[\boldsymbol{\rho}(\ln\boldsymbol{\rho} - \ln\boldsymbol{\sigma})]$$

Network thresholding

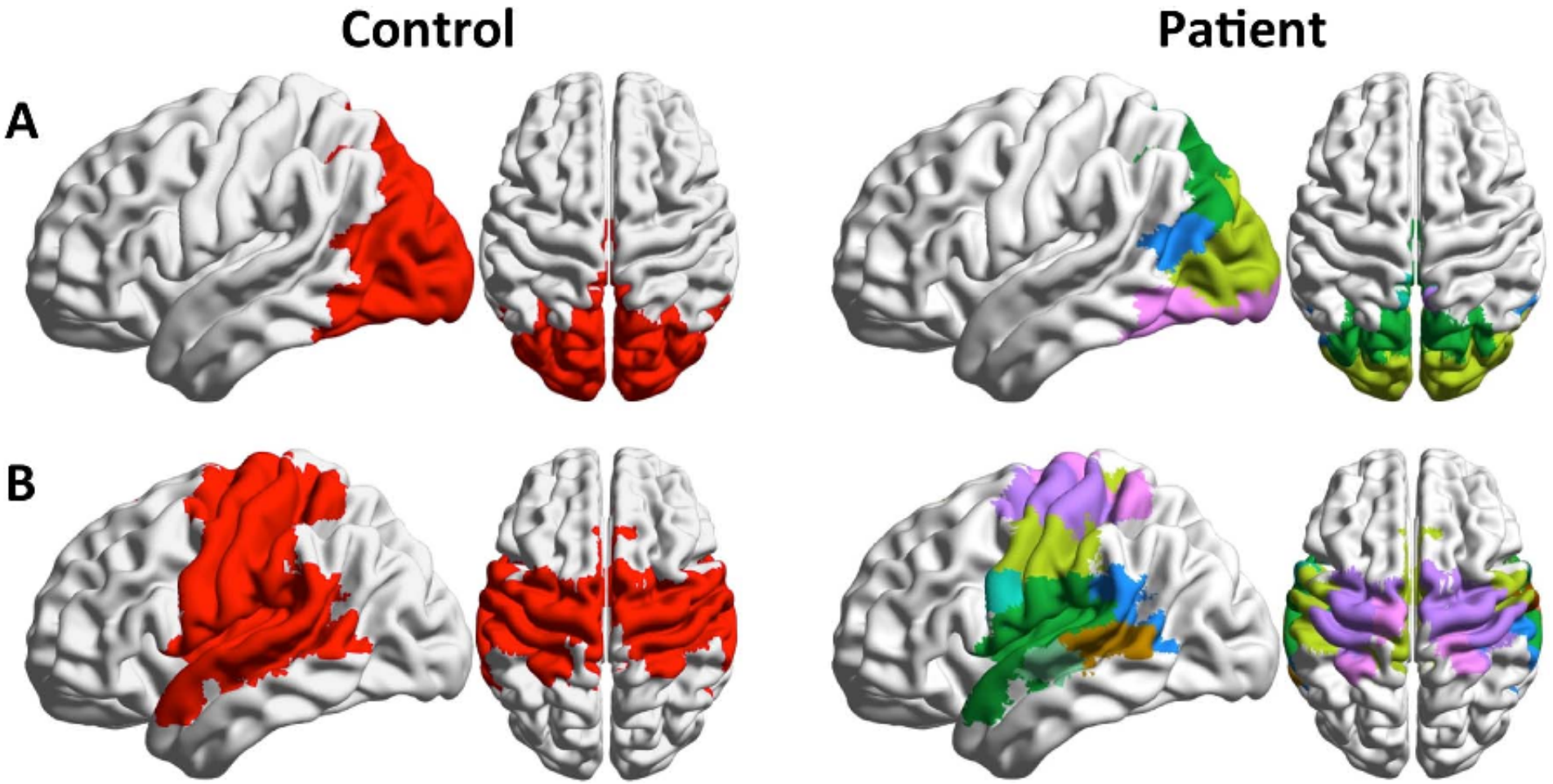
- Sparsification to reduce effects of experimental noise.
- Computationally tractable.
- rs-fMRI networks, N=638, 24 healthy subjects.



Effects of network thresholding



Disruption of modular architecture in schizophrenia



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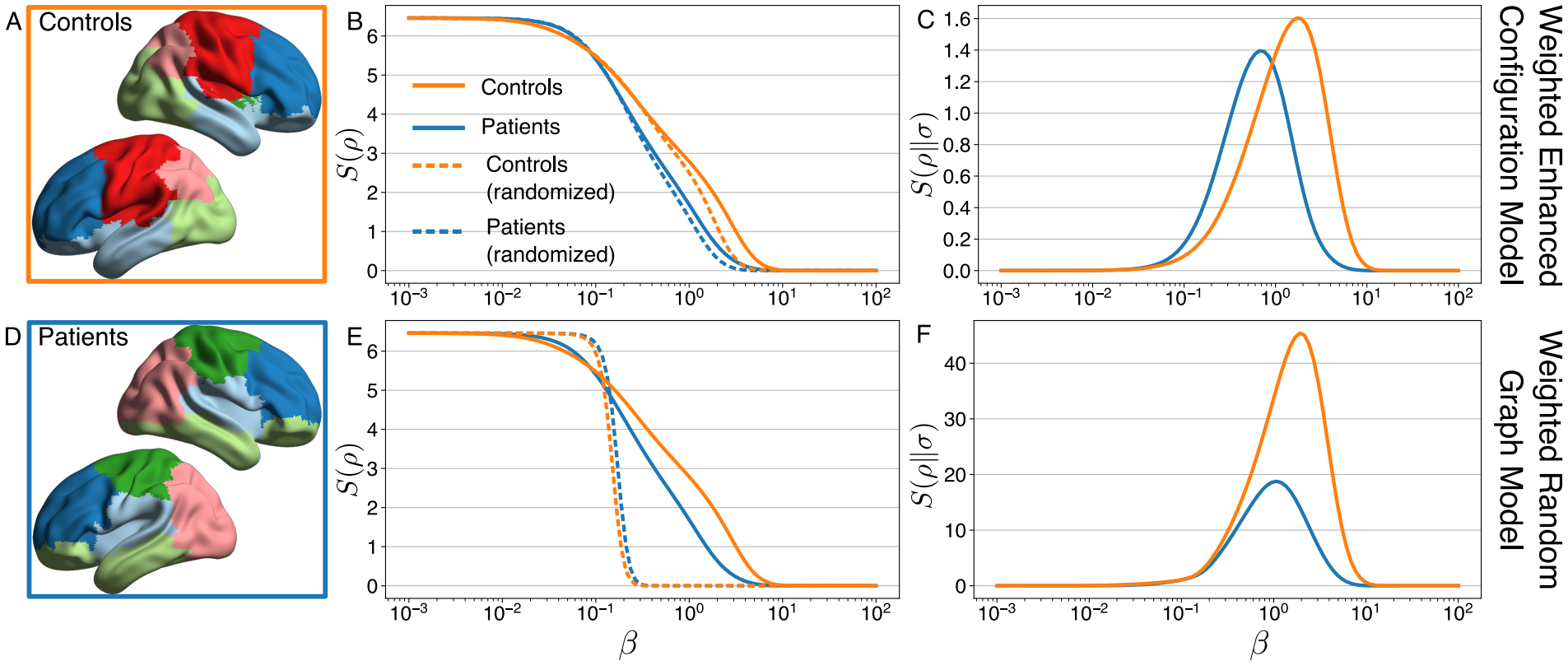
[1]Bordier et al. Neuroimage: Clinical (2018)

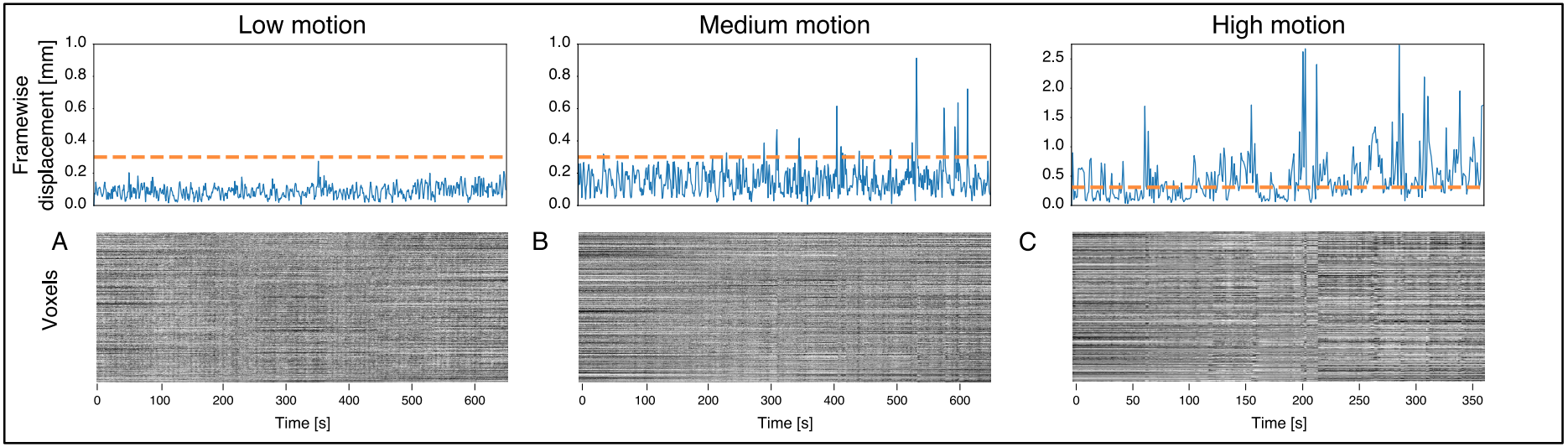
Effects of disease: schizophrenia

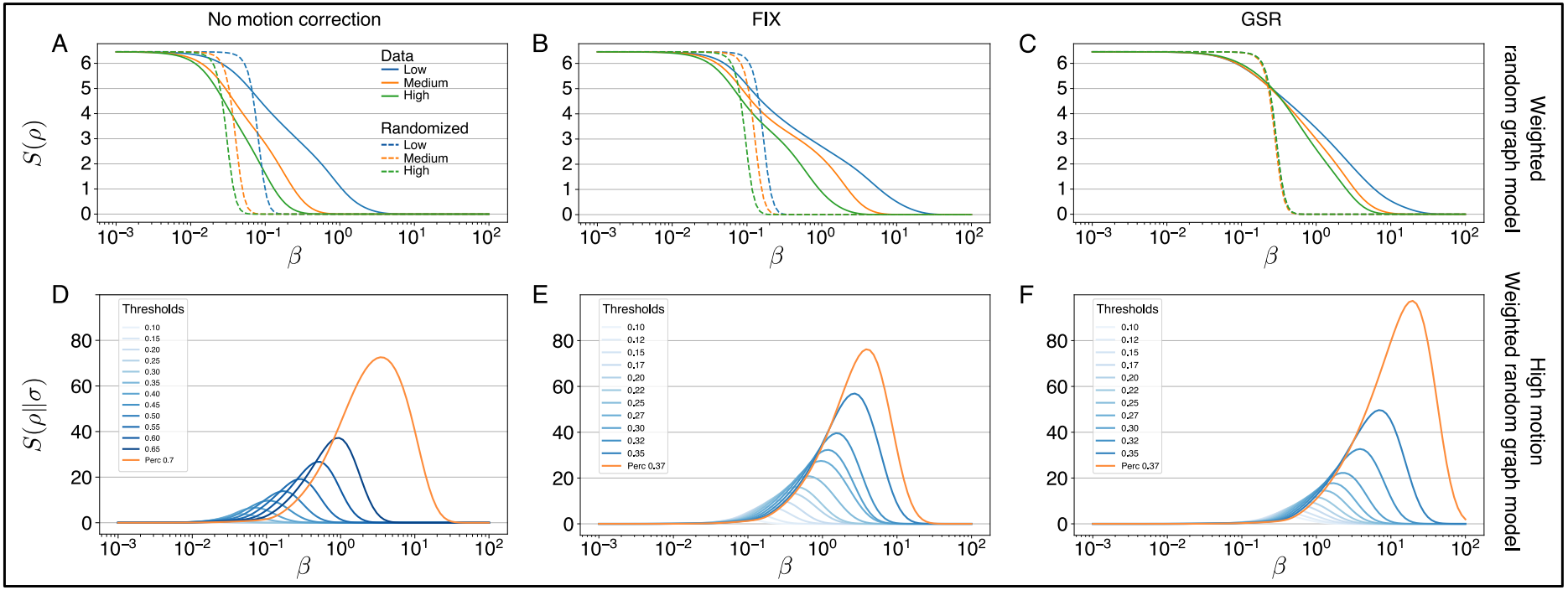


Motion and its correction

- Long range spurious correlation
- Pipelines for correction
- Aggressive vs. conservative

Effects of motion





Conclusions

- rs-fMRI null models
- x
- y